Version 7.2.11

PVsyst - Simulation report

Pumping PV System

Project: GreenField Irrigation Solar Pumping

Variant: Solar Irrigation Project

Pumping PV System

System power: 450 Wp

Ngata - Kenya

Author

PVsyst V7.2.11
VC0, Simulation date:
17/02/26 17:45
with v7.2.11

Project summary

Geographical Site
Ngata
Kenya

Situation

Latitude	-0.26 °S
Longitude	36.01 °E
Altitude	1977 m
Time zone	UTC+3

Project settings

Albedo	0.21

Meteo data
Ngata
NASA-SSE satellite data 1983-2005 - Synthetic

System summary

Pumping PV System
Simulation for year no 10

Deep Well to Storage

PV Field Orientation

Fixed plane	
Tilt/Azimuth	11 / 0 °

Near Shadings
Linear shadings

Water needs

Yearly constant	10.00 m³/day

System information
PV Array

Nb. of modules	3 units
Pnom total	450 Wp

Results summary

Water

Water Pumped	3062 m³
Specific	2892 m³/kWp/bar
Water needs	3650 m³
Missing Water	16.1 %

Energy

Energy At Pump	621 kWh
Specific	0.20 kWh/m³
Unused (tank full)	
Unused PV energy	10 kWh
Unused Fraction	1.5 %

Efficiencies

System efficiency	90.4 %
Pump efficiency	32.6 %

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General parameters

Pumping PV System

System Requirements

Basic Head	23 meterW

Water needs

Yearly constant	10.00 m³/day

Deep Well to Storage

Well characteristics

Static level depth	-15.0 m
Specific drawdown	-1.00 m/m³/h
Diameter	15 cm
Pump level	-30.0 m
Lower dynamic level	-20.0 m

Storage tank

Volume	15.0 m³
Diameter	2.5 m
Feeding by top	
Feeding altitude	8.0 m
Height (full level)	3.0 m

Hydraulic circuit

Piping length	50 m
Pipes	PE50
Dint	54 mm
Number of elbows	3
Other friction losses	7.000

PV Field Orientation

Fixed plane	
Tilt/Azimuth	11 / 0 °

Near Shadings

Linear shadings

PV Array and Pump

PV module

Manufacturer	Yangzhou Tianhua
Model	SDM-170/150-72M
(Original PVsyst database)	
Unit Nom. Power	150 Wp
Number of PV modules	3 units
Nominal (STC)	450 Wp
Modules	3 Strings x 1 In series
At operating cond. (50°C)	
Pmpp	397 Wp
U mpp	30 V
I mpp	13 A
Total PV power	
Nominal (STC)	0 kWp
Total	3 modules

Pump

Manufacturer	All Power CSV
Model	Watermax PA
Pump Technology	Membrane/Diaphragm
	Deep well pump
Motor	DC motor, brushless

Operating conditions

	Head min.	Head Nom	Head max.	
	10.0	60.0	80.0	m
Corresp. Flowrate	1.52	1.30	1.20	m³/h
Req. power	200	470	565	W

Control device

Model	Generic device (optimised for the system)
System Configuration	MPPT-DC converter

Pumping system controller

System Operating Control

Generic device params adjusted acc. to the system

Power Conditioning Unit

Type	MPPT-DC converter

Operating conditions

Nominal power	570 W
Power Threshold	6 W
Max. efficiency	97.0 %
EURO efficiency	95.0 %
Minimum MPP Voltage	20 V
Maximum MPP Voltage	50 V
Maximum Array Voltage	52 V
Maximum Input Current	9.0 A

System losses

Array Soiling Losses
Loss Fraction 3.0 %

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 29.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 40 mΩ
Loss Fraction 1.5 % at STC

Serie Diode Loss
Voltage drop 0.7 V
Loss Fraction 2.0 % at STC

LID - Light Induced Degradation
Loss Fraction 2.0 %

Module Quality Loss
Loss Fraction 1.5 %

Module mismatch losses
Loss Fraction 2.0 % at MPP

Strings Mismatch loss
Loss Fraction 0.1 %

Module average degradation
Year no 10
Loss factor 0.4 %/year
Mismatch due to degradation
Imp RMS dispersion 0.4 %/year
Vmp RMS dispersion 0.4 %/year

IAM loss factor
Incidence effect (IAM): Fresnel smooth glass, n = 1.526

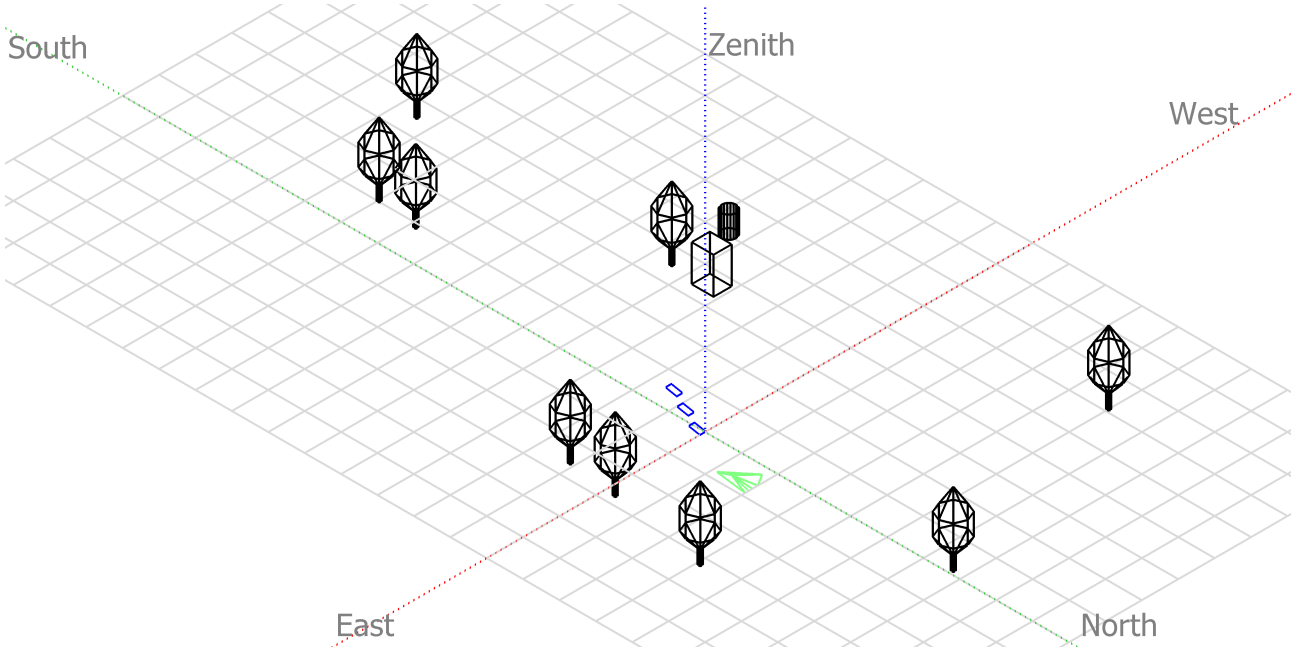
0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.403	0.000

Spectral correction
FirstSolar model
Precipitable water estimated from relative humidity

Coefficient Set	C0	C1	C2	C3	C4	C5
Monocrystalline Si	0.85914	-0.02088	-0.0058853	0.12029	0.026814	-0.001781

Near shadings parameter

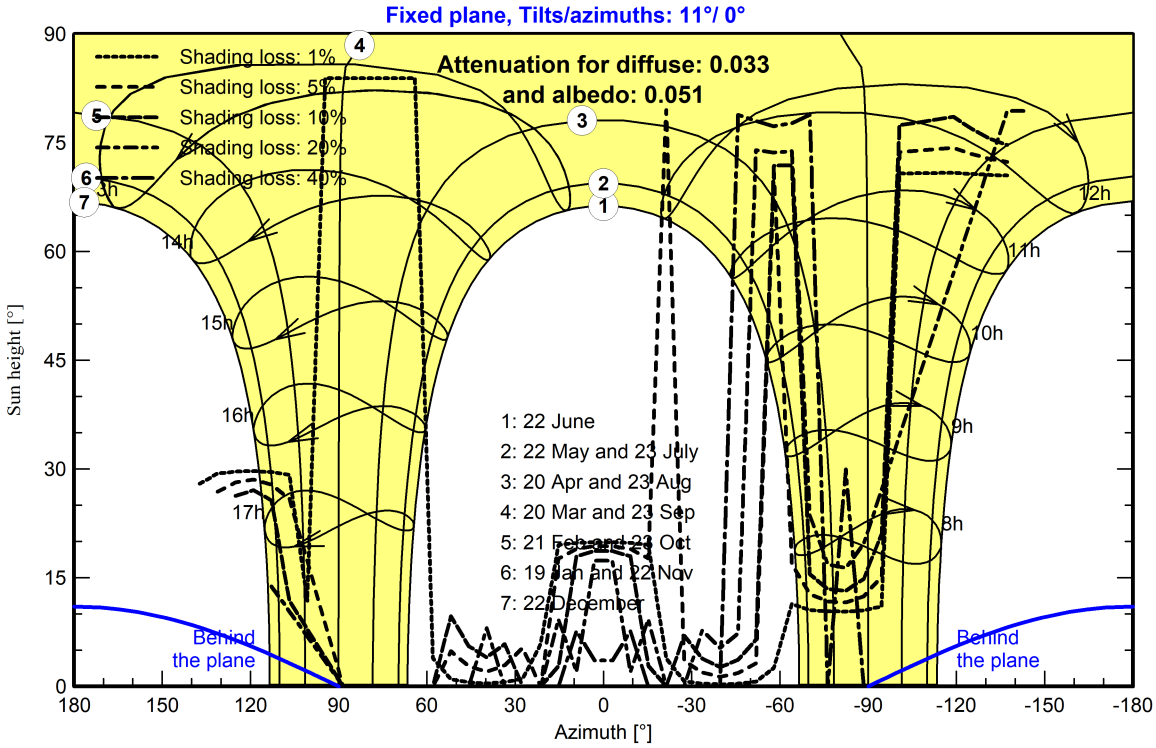
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 11°/ 0°



Main results

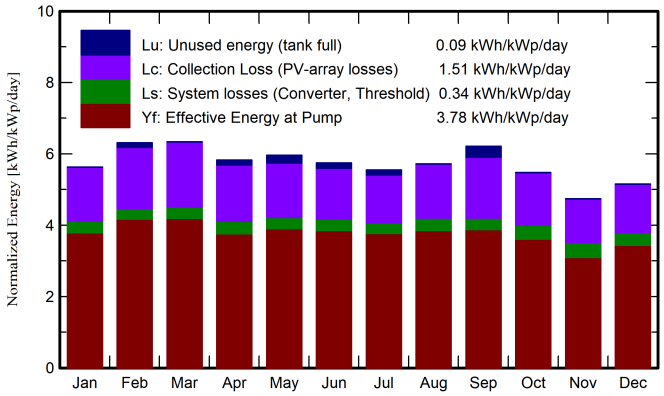
System Production

Water		Energy		Efficiencies	
Water Pumped	3062 m³	Energy At Pump	621 kWh	System efficiency	90.4 %
Specific	2892 m³/kWp/bar	Specific	0.20 kWh/m³	Pump efficiency	32.6 %
Water needs	3650 m³	**Unused (tank full)**			
Missing Water	16.1 %	Unused PV energy	10 kWh		
		Unused Fraction	1.5 %		

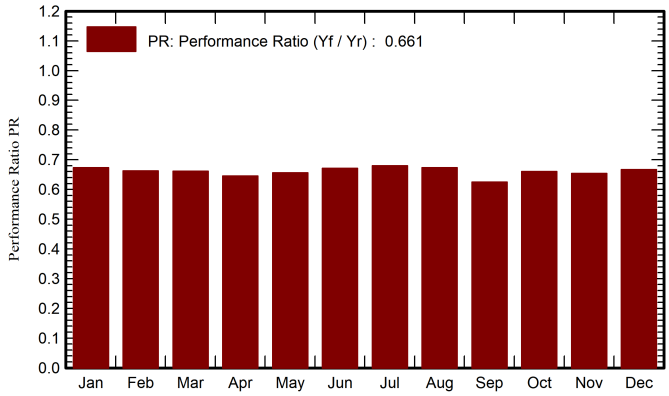
Economic evaluation

Investment		Yearly cost		Specific Cost	
Global	2,934.00 USD	Annuities	0.00 USD/yr	Water Cost	0.01 USD/m³
Specific	6.52 USD/Wp	Run. costs	66.13 USD/yr	Energy cost	0.06 USD/kWh

Normalized productions (per installed kWp)



Performance Ratio PR



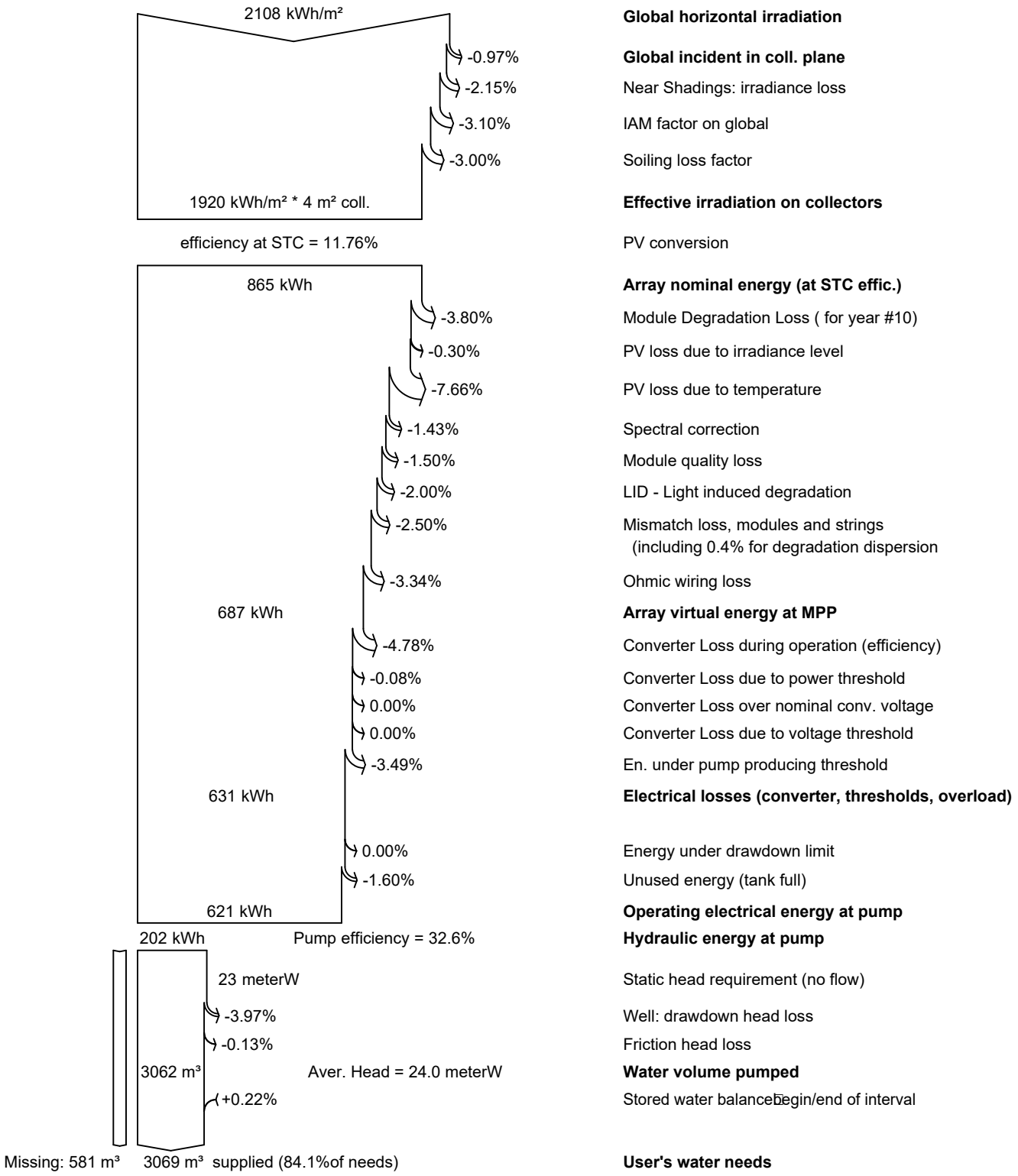
Balances and main results

	GlobEff	EArrMPP	E_PmpOp	ETkFull	H_Pump	WPumped	W_Used	W_Miss
	kWh/m²	kWh	kWh	kWh	meterW	m³	m³	m³
January	160.9	57.52	52.93	0.000	24.02	264.1	261.6	48.36
February	162.7	57.48	52.67	1.056	24.06	263.6	270.4	9.55
March	178.7	63.06	58.56	0.000	24.06	293.7	289.3	20.68
April	159.9	56.87	50.80	1.340	23.97	249.7	250.2	49.77
May	171.5	61.14	54.56	2.151	23.97	266.7	268.4	41.55
June	161.3	57.90	52.06	1.438	24.03	258.3	259.4	40.64
July	160.2	58.12	52.56	1.275	23.93	256.2	260.3	49.66
August	163.1	58.68	53.74	0.000	23.98	264.7	257.2	52.78
September	168.3	59.70	52.43	2.861	23.98	256.4	262.9	37.06
October	156.3	55.87	50.43	0.000	23.98	248.2	248.9	61.12
November	130.7	47.53	41.89	0.000	23.89	204.2	201.3	98.67
December	146.4	53.01	48.02	0.000	23.93	236.2	238.4	71.57
Year	1919.7	686.88	620.64	10.121	23.98	3061.9	3068.6	581.40

Legends

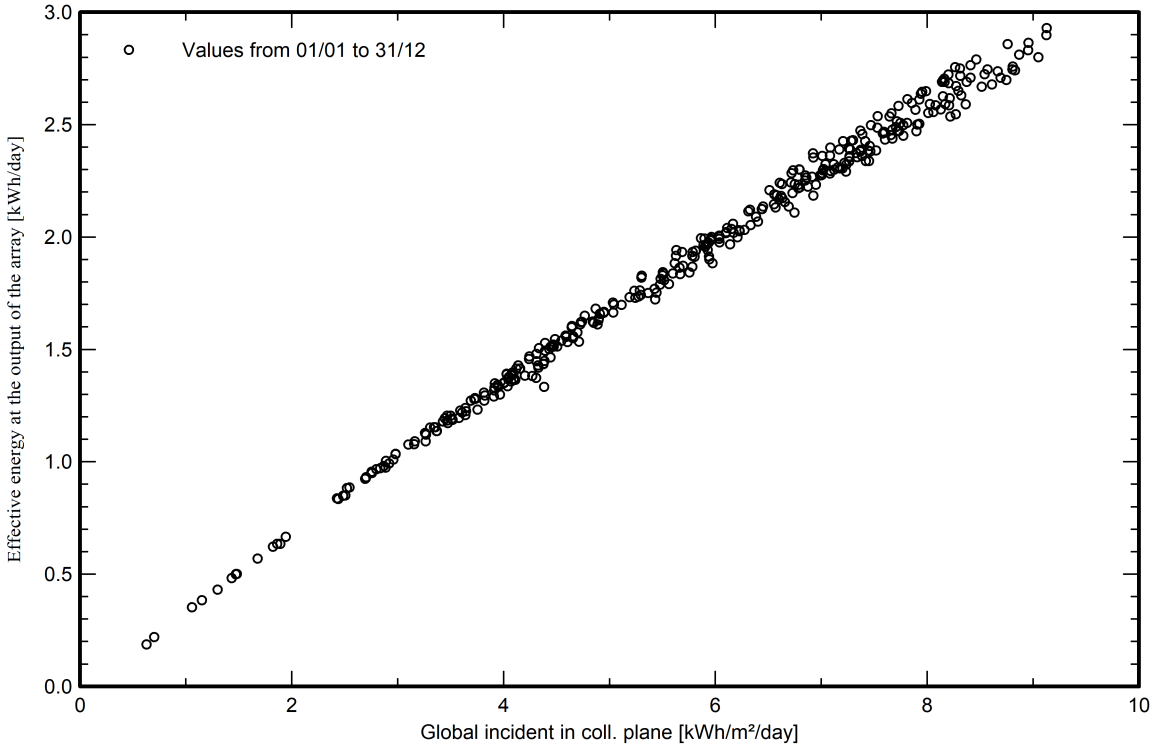
GlobEff	Effective Global, corr. for IAM and shadings	WPumped	Water volume pumped
EArrMPP	Array virtual energy at MPP	W_Used	Water drawn by the user
E_PmpOp	Pump operating energy	W_Miss	Missing water
ETkFull	Unused energy (tank full)		
H_Pump	Average total Head at pump		

Loss diagram



Special graphs

Daily Input/Output diagram



Cost of the system

Installation costs

Item	Quantity units	Cost USD	Total USD
Pumps			272.00
Controllers			85.00
Tank			527.00
Hydraulic circuit	9	10.00	90.00
Other components			
Accessories, fasteners	1	310.00	310.00
Wiring	1	129.00	129.00
Combiner box	1	39.00	39.00
Monitoring system, display screen	1	39.00	39.00
Measurement system, pyranometer	1	15.00	15.00
Surge arrester	1	39.00	39.00
Installation			
Global installation cost per module	3	463.00	1,389.00
Total			2,934.00
Depreciable asset			667.00

Operating costs

Item	Total USD/year
Maintenance	
Salaries	40.00
Total (OPEX)	40.00
Including inflation (5.00%)	66.13

System summary - Water and Energy cost

Total installation cost	2,934.00 USD
Operating costs (incl. inflation 5.00%/year)	66.13 USD/year
Energy used for pumping	621 kWh/year
Excess energy (tank full)	10.1 kWh/year
Water Pumped	3062 m³
Cost of pumped water	0.07 USD/m³